Master's Thesis

Suppression of leakage via pulse waveform optimization in superconducting quantum systems

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Chapter 1 Introduction

1.1 Background and Motivation

Quantum computing represents a paradigm shift in computational science, offering exponential speedups for specific classes of problems, including quantum chemistry simulation, cryptography, and large-scale optimization. At the heart of this physical realization are quantum bits (qubits). Among the various physical implementations, superconducting qubits-particularly the transmon architecture-have emerged as one of the most promising platforms. Their macroscopic nature allows for strong coupling to microwave control fields and flexible lithographic fabrication, making them highly scalable.However, realizing fault-tolerant quantum computation requires extremely high-fidelity quantum operations. Current Noisy Intermediate-Scale Quantum (NISQ) devices are fundamentally limited by decoherence and operational errors. For superconducting qubits, single- and two-qubit gate fidelities are highly sensitive to the precise shape of the microwave control pulses used to drive the transitions. Imperfect control pulses lead to phase errors, over-rotation, and leakage out of the computational subspace (e.g., transitions into the $|2\rangle$ state), severely degrading the overall performance of the quantum processor.1.2 Problem StatementThe transmon qubit is inherently a weakly anharmonic oscillator. Because the energy difference between the $|0\rangle \leftrightarrow |1\rangle$ transition and the $|1\rangle \leftrightarrow |2\rangle$ transition is relatively small, applying fast, high-amplitude microwave pulses to minimize decoherence time inevitably results in spectral leakage into higher, non-computational energy states.While conventional approaches such as the Derivative Removal by Adiabatic Gate (DRAG) technique have successfully suppressed this leakage to a large extent, the demand for even higher fidelities necessitates more advanced waveform engineering. Standard Gaussian or square pulses are often spectrally broad or susceptible to noise. There is a critical need to explore mathematically optimal pulse shapes-such as Slepian pulses-that can strictly confine the energy spectrum while maintaining fast gate times, and to develop robust calibration frameworks to validate these pulses against realistic Hamiltonian dynamics.1.3 Objectives and ContributionsThe primary objective of this thesis is to investigate, optimize, and benchmark advanced gate pulse waveforms for superconducting transmon qubits. By applying numerical optimal control and advanced signal processing techniques, this work aims to minimize gate errors induced by spectral leakage and environmental noise.The specific contributions of this thesis are as follows:Pulse Waveform Engineering: Development and numerical evaluation of advanced pulse shapes, including DRAG-corrected Slepian pulses, to tightly bound the frequency spectrum of the control signals.System Simulation: Implementation of open quantum system dynamics to model a transmon qubit, utilizing time-dependent Hamiltonians to accurately capture the effects of the applied microwave waveforms.Gate Calibration: Establishment of a robust numerical calibration routine to optimize pulse parameters (e.g., amplitude, detuning, and duration) to maximize gate fidelity.

1.2 Thesis Organization

The remainder of this thesis is structured to guide the reader from the fundamental physics of the system through the mathematical modeling and finally to the numerical results of the optimized waveforms.

Chapter 2 Theoretical Foundations: Single-Qubit X Gate

After we briefly overview the basics of quantum bits, this chapter introduces some fundamental theoretical frameworks and reviews prior methodologies essential for understanding the optimization of control pulses in superconducting quantum systems. The Derivative Removal by Adiabatic Gate (DRAG) formalism, Schrieffer-Wolff transformations, and Riccati engineering serve as the baseline for the advanced waveform optimizations proposed in this study. In exploring these control methodologies, we first consider the case of the one-qubit X gate before later extending our analysis to the two-qubit CZ gate architecture.

2.1 Introduction to the Quantum Bit

The transition from classical to quantum information processing represents a fundamental paradigm shift in computational theory. In classical computing, the foundational unit of information is bit which exists in a mutually exclusive, deterministic state of either 0 or 1. Quantum computing replaces this discrete entity with the quantum bit, or qubit, an entity governed by the principles of quantum mechanics. Rather than being confined to a single state, a qubit can exist in a coherent superposition of both states simultaneously, unlocking exponentially larger state spaces and enabling computational speedups for specific classes of problems, such as integer factorization and molecular simulation.

Mathematically, the state of a single, isolated qubit is represented as a state vector in a two-dimensional complex Hilbert space. Using Dirac notation, the computational basis states are denoted as $|0\rangle$ and $|1\rangle$, corresponding to the classical 0 and 1. A pure qubit state $|\psi\rangle$ is expressed as a linear combination of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$. This constraint allows the pure state of a qubit to be mapped onto the surface of a unit sphere in three-dimensional space, known as the Bloch sphere. On the Bloch sphere, the poles represent the basis states $|0\rangle$ and $|1\rangle$, while points on the equator represent equal superpositions with varying relative phases. This geometric representation is indispensable for visualizing single-qubit quantum gates as rotations about the sphere's axes.

As a quantum mechanical system, a qubit's dynamics are governed by the time-dependent Schrödinger equation. Consequently, any control applied to the system is mathematically represented by the Hamiltonian operator driving this evolution. In formula, [15]

$$\psi(t) = \mathcal{T}\left(\exp\left[\int_0^T iH(t)\,\mathrm{d}t\right]\right)\psi(0).$$

Here, $\mathcal{T}$ denotes the Dyson time-ordering operator, T is the total gate duration, and $H(t)$ is the system's total Hamiltonian. In practice, $H(t)$ consists of a static drift term defining the unperturbed system and a time-dependent control term corresponding to external driving fields, such as shaped microwave pulses. The primary objective of quantum control is to engineer these time-dependent waveforms so that the resulting evolution operator accurately synthesizes a desired target unitary gate on the computational subspace.

2.2 Gate Fidelity and Leakage

While the idealized model assumes a perfect two-level system, many physical implementations, notably transmon qubits, are not strictly confined to two states. For instance, weakly anharmonic oscillators possess a ladder of energy levels where the computational subspace consisting of $|0\rangle$ and $|1\rangle$ is not entirely isolated from higher-excited states, most notably the $|2\rangle$ state.When applying short, high-bandwidth microwave pulses to execute fast quantum operations, the broad frequency spectrum of the control signal can inadvertently drive transitions into these non-computational states. The phenomenon where population escapes the $\{|0\rangle, |1\rangle\}$ subspace is known as leakage. Because standard quantum error correction codes are primarily designed to handle errors within the computational basis, leakage represents a particularly damaging source of information loss.To mathematically quantify the success of a control pulse, we evaluate the gate fidelity, which measures the closeness of the actual physical evolution to the ideal target unitary operation. Leakage fundamentally limits this metric; any population that leaves the computational subspace immediately degrades the fidelity of the operation.Consequently, achieving the high-fidelity gates required for fault-tolerant quantum computing relies heavily on advanced waveform optimization. The control fields defined in $H(t)$ must be meticulously shaped to strike a delicate balance: driving the desired rotations on the Bloch sphere as rapidly as possible to avoid decoherence, while simultaneously suppressing the spectral components that cause leakage to higher energy levels.

2.3 Qudit Hamiltonian in the Rotating Frame

As noted in 2.2, A transmon qubit is a widely used superconducting qubit. Because a transmon is inherently an anharmonic oscillator, it possesses higher excited states beyond $|0\rangle$ and $|1\rangle$. Consequently, it acts as a qudit—a quantum system with d levels—rather than an ideal qubit, which inevitably introduces leakage into unwanted noncomputational states.

To model this leakage analytically, we review the derivation of the qudit Hamiltonian truncated at the first $N_{\max}+1$ levels of the transmon by following Ref. [8]. In passing, we derive the Hamiltonian without the Rotating Wave Approximation (RWA)[3].

The Hamiltonian of a transmon qudit in the lab frame is [16]

$$\hat{H}_{\text{lab}} = 4E_c(\hat{n} - \hat{n}_g)^2 - E_J \cos(\hat{\varphi}), \tag{1}$$

where $\hat{n}_g$ denotes the offset charge, $\hat{n}$ the number operator, E_C the charging energy, E_J the Josephson energy, and φ the phase operator. From the (non-degenerate) perturbation theory, we find that the eigenstates of this Hamiltonian are harmonics with non-uniform energy spacings. We denote the k-th excited state as $|k\rangle$ and its φ-representation $\psi_k(\varphi)$. Now we consider the expansion of $\hat{n}$ in $|k\rangle\langle m|$ ($k \geq 0$, $m \geq 0$). Note that k's parity coincides with ψ_k's. Since $\hat{n}$ flips the sign, the coefficient

$$\lambda_{k,m} = \langle k|\,\hat{n}\,|m\rangle \;=\; \int_{-\pi}^{\pi} \psi_k^*(\varphi)\,(-i) \tag{2}$$

is zero unless k and m are of opposite parity. Hence $\hat{n}$ can be expressed as

$$\begin{aligned}\hat{n} &= \sum_{k=0}^{N_{\max}-1} \Big[\lambda_{k,k+1}|k\rangle\langle k+1| \\ &+ \sum_{j=1} \lambda_{k,k+2j+1}|k\rangle\langle k+2j+1|\Big] + \text{h.c.}\end{aligned} \tag{3}$$

If we apply the change-of-basis V on the Hilbert space, the Hamiltonian changes as

$$H' = VHV^\dagger + i\dot{V}V^\dagger, \tag{4}$$

where the last term is called a gauge term. Applying the diagonal frame transformation with

$$R^\dagger = \exp\left(i\omega_d t \sum_{k=0}^{N_{\max}} k|k\rangle\langle k|\right) \tag{5}$$

yields the rotating frame Hamiltonian

$$\begin{aligned}\hat{H}_{\text{rot}} = \sum_{k=0}^{N_{\max}} \Delta_k\,|k\rangle\,\langle k| + \Omega_{\text{RF}}(t)\Big[\\ \sum_{k=0}^{N_{\max}-1} \sum_{j=1} \lambda_{k,k+2j+1}|k\rangle\langle k+2j+1|e^{-(2j+1)i\omega_d t} \\ + \sum_{k=0}^{N_{\max}-1} \lambda_{k,k+1}|k\rangle\langle k+1|e^{-i\omega_d t} + \text{h.c.}\Big],\end{aligned} \tag{6}$$

where $\Delta_k = \omega_k - k\omega_d$, ω_d is the drive frequency and $\omega_k = E_k - E_0$. Henceforth, if we simply write Δ, this means Δ_2.

If we apply RWA entirely, Eq. (6) reduces to

$$\begin{aligned}\hat{H}_{\text{RWA}} &= \sum_{j=1}^{N_{\max}} (\Delta_j + j\delta(t))\,|j\rangle\,\langle j| \\ &+ \sum_{j=1}^{N_{\max}} \lambda_j\left[\frac{\Omega_x(t)}{2}\hat{\sigma}^x_{j-1,j} + \frac{\Omega_y(t)}{2}\hat{\sigma}^y_{j-1,j}\right],\end{aligned} \tag{7}$$

where $\lambda_j = \lambda_{j-1,j}$, $\hat{\sigma}^x_{j-1,j} = |j-1\rangle\,\langle j| + |j\rangle\,\langle j-1|$, and $\hat{\sigma}^y_{j-1,j} = -i\,|j-1\rangle\,\langle j| + i\,|j\rangle\,\langle j-1|$.

The matricial form when $N_{\max} = 3$ is

$$\begin{pmatrix} 0 & \frac{1}{2}\lambda_1(\Omega_x(t) - i\Omega_y(t)) & 0 & 0 \\ \frac{1}{2}\lambda_1(\Omega_x(t) + i\Omega_y(t)) & \delta(t) + \Delta_1 & \frac{1}{2}\lambda_2(\Omega_x(t) - i\Omega_y(t)) & 0 \\ 0 & \frac{1}{2}\lambda_2(\Omega_x(t) + i\Omega_y(t)) & 2\delta(t) + \Delta_2 & \frac{1}{2}\lambda_3(\Omega_x(t) - i\Omega_y(t)) \\ 0 & 0 & \frac{1}{2}\lambda_3(\Omega_x(t) + i\Omega_y(t)) & 3\delta(t) + \Delta_3 \end{pmatrix}. \tag{8}$$

Including the first order corrections of RWA, the Hamiltonian becomes

$$\begin{aligned}\hat{H}_{\mathrm{RWA},1} = \sum_{k=0}^{N_{\max}} \tilde{\Delta}_k \left|k\right\rangle \left\langle k\right| + \frac{1}{2}\left(\Omega_x - i\Omega_y\right)\Bigg[\\ \sum_{k=0}^{N_{\max}-1} \lambda_{k,k+3} |k\rangle\langle k+3| e^{-2i\omega_d t} \\ + \sum_{k=0}^{N_{\max}-1} \lambda_{k,k+1} |k\rangle\langle k+1| \Bigg] + \text{h.c.}\end{aligned} \tag{9}$$

where, for brevity, the last h.c. only refers off-diagonal components.

In this study, we consider up to the second excited state in our analysis.

2.4 Approximation of λ

In the QHO approximation, $\hat{n}$ is expressed in terms of the annihilation and creation operators:

$$\hat{n} \approx i n_{\mathrm{zpf}} (\hat{a}^\dagger - \hat{a})$$

Recall $\hat{a} = \sum_{k=0}^{\infty} \sqrt{k+1} \left|k+1\right\rangle \left\langle k\right|$.

$$\left\langle k+1\right| \hat{a} \left|k\right\rangle = \sqrt{k+1} \quad \left\langle k+1\right| \hat{a}^\dagger \left|k\right\rangle = 0$$

Hence,

$$\lambda_{k,k+1} \approx \left\langle k+1\right| \hat{n} \left|k\right\rangle = i n_{\mathrm{zpf}} \sqrt{k+1}$$

The prefactor $i n_{\mathrm{zpf}}$ could be regarded as absorbed by the drive amplitude $\Omega_{\mathrm{RF}}(t)$. Hence we simply set

$$\lambda_{k,k+1} \approx \sqrt{k+1}.$$

For λ_1 is entirely aborbed into $\Omega_{\mathrm{RF}}(t)$, it is exactly (not approximately) $\lambda_1 = 1$.

In simulations, we set the values of λ accordiingly.

2.5 Schrieffer-Wolff Transformation

While the rotating frame tranformation in Eq. (6) is computed exactly, such exact analytical transformations are not always achievable. In general, the generator S is chosen so that $i\dot{S}$ cancels the error term in the Hamiltonian. In Eq. (4), V is assumed to be a unitary operator. we now consider a scenario where the transformation operator V is parameterized by an anti-Hermitian operator S. Specifically, V is defined exponentially as:

$$V = \exp(S),$$

we can expand the Hamiltonian in a power series of S as

$$H' = i\dot{S} + H + [S, H] + \frac{1}{2}[S, [S, H]] + \cdots . \quad \text{(Hadamard's lemma)} \tag{10}$$

It is important to emphasize that this expansion is mathematically rigorous and holds generally true for any generator S and Hamiltonian H, provided the system is defined within a finite-dimensional Hilbert space. While truncating the series early causes inaccuracies, adding more terms makes subsequent computations excessively complex. In Chapter 4, we develop a method to the approximation is valid.

2.6 Derivative Removal by Adiabatic Gate (DRAG)

The Derivative Removal by Adiabatic Gate (DRAG) technique is a highly effective method for suppressing leakage. It works by applying a secondary corrective pulse–proportional to the time derivative of the primary pulse envelope–along an orthogonal control axis, which actively cancels unwanted transitions to higher energy levels (such as the $|2\rangle$ state in transmon qubits) and corrects phase errors. Gambetta and Motzoi [4] derived the DRAG Y correction formula

$$\Omega_y = -\frac{1}{\Delta}\dot{\Omega}_x, \tag{11}$$

and demonstrated that that AC Stark shift can be compensated by way of phase ramping,

$$\delta(t) = \frac{4 - \lambda^2}{4\Delta}\,\Omega_x^2. \tag{12}$$

Collectively Eq. (11) and Eq. (12) are referred to as DRAG conditions. When these two conditions are applied within the weak driving regime, the Hamiltonian becomes

$$\hat{\mathcal{H}}_1(t) \approx \frac{\Omega_x}{2}\hat{\sigma}^x_{0,1} + \frac{\lambda_2\Omega_x^2}{8\Delta_2}\hat{\sigma}^x_{0,2} - \Omega_x^2\left[\frac{(2\Delta_2 - \Delta_3)}{8\Delta_2^2}\right]\lambda_2\lambda_3\hat{\sigma}^x_{1,3}, \tag{13}$$

subject to the following boundary conditions:

$$\Omega_x(t) = \dot{\Omega}_x(t) = 0 \quad \text{at } t = 0,\, T.$$

2.6.1 The DRAG-1 Scheme

Hamiltonian Eq. (13) is remained with two photon transitions between $|0\rangle \leftrightarrow |2\rangle$ and $|1\rangle \leftrightarrow |3\rangle$. SWTs to further reduce these transitions is applicable. Now we introduce a recursive DRAG method called DRAG-1 [10] or R1D [8]:

Applying Schrieffer-Wolff transformations with $S_2 = -i\lambda_2\Omega_1^2/(8\Delta_2^2)\hat{\sigma}^y_{0,2}$ and $S_3 = i\lambda_2\dot{\Omega}_1\Omega_1/(4\Delta_2^3)\hat{\sigma}^x_{0,2}$ to Eq. (13) yields

$$\hat{\mathcal{H}}_{\text{new}}(t) = \frac{\Omega_x}{2}\hat{\sigma}^x_{0,1} + \frac{\lambda_2\Omega_x^2}{8\Delta_2}\hat{\sigma}^x_{0,2} - \Omega_x^2\left[\frac{(2\Delta_2 - \Delta_3)}{8\Delta_2^2}\right]\lambda_2\lambda_3\hat{\sigma}^x_{1,3} - \lambda_2\left[\right]\sigma^x_{0,2}$$

Therefore,

$$\Omega_x(t) = \sqrt{\Omega_1^2 + \frac{2}{\Delta_2^2}(\dot{\Omega}_1^2 + \ddot{\Omega}_1\Omega_1)} \tag{14}$$

Ω_1 has to satisfy the following boundary conditions

$$\Omega_1 = 0 \quad \text{at } t = 0,\, T$$

In case of X gate, the X gate condition

$$\int_0^T \Omega_x(t)\,\mathrm{d}t = \pi \tag{15}$$

gets a slight modification if we consider up to the third degree in the frame transformation Eq. (27)

$$\Omega_x'(t) = \Omega_x(t) - \frac{(4-\lambda_2^2)\Omega_x^3(t)}{8\Delta^2}.$$

(Corrections of this sort in the DRAG scheme are called superlinear corrections.)

2.7 NPAD

A lot of efforts for leakage reduction and fidelity improvements have been done in the perturbative regime (i.e. $\Omega/\Delta \ll 1$). One of the framework advocated for the non-perturbative strong driving regime is NPAD [11]. NPAD is an iterative method to diagonalize a nonperturbative Hamiltonian by applying static Givens rotations until the desired accuracy is achieved. While this method is only effective if the Hamiltonian has a rather simple form and only applicable to time-independent Hamiltonian, theoretically this is a generalization of recursive SWT in the perturbative regime.

2.8 Block Diagonalization via Riccati Engineering

Although widely known as a qunatum control method, Riccati engineering is not yet applied to time-dependent one qubit gate controls. (Time-independent cases are investigated in [5, 6].) Given a Hamiltonian

$$H(t) = \begin{pmatrix} H_{11}(t) & H_{12}(t) \\ H_{21}(t) & H_{22}(t) \end{pmatrix},$$

we consider block diagonalization of $H(t)$ into 2×2 and 1×1 partitions according to the transformation Eq. (4).

The motivation is that if we could block diagonalize H in this way, the dynamics starting with $|0\rangle$ in this new frame will cause zero leakage into the non-computational subspace. As long as the off-diagonal terms are zero. The solution to this problem is given in [6],

$$U(t) = \begin{pmatrix} (I_2 + X^\dagger X)^{-1/2} & X^\dagger (I_{n-2} + XX^\dagger)^{-1/2} \\ -X(I_2 + X^\dagger X)^{-1/2} & (I_{n-2} + XX^\dagger)^{-1/2} \end{pmatrix}$$

with $X(t) = (x_1(t) \quad x_2(t))$ satisfying

$$i\dot{X} = H_{21} + H_{22}X - XH_{11} - XH_{12}X \tag{16}$$

Under this frame change, the resulting $H'_{11}(t)$ is given by

$$H'_{11}(t) = \left(I_2 + X^\dagger X\right)^{1/2} (H_{11}+H_{12}X)(I_2+X^\dagger X)^{-1/2} - (I_2+X^\dagger X)^{1/2}\partial_t \left[(I_2 + X^\dagger X)^{-1/2}\right] \tag{17}$$

The form of this Hamiltonian is not very easy to analyze.

2.9 Issues with the Current Research

Chapter 3 Theoretical Foundations: Two-Qubit CZ gate

In this section, we overview start dealing with the two qubit CZ gate. Note that we use different notations for X and CZ gates.

3.1 Basics

CZ gate is a two-qubit gate that applies a phase flip to the target qubit if the control qubit is in the state $|1\rangle$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the ideal unitary evolution is represented by a diagonal matrix:

$$U_{\mathrm{CZ}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{18}$$

3.2 SQUID loop and Z-control

In superconducting quantum architectures, the ability to dynamically tune the transition frequency of a qubit is essential for realizing the adiabatic CZ gate. This tunability is achieved by replacing the single Josephson junction of a standard transmon with a superconducting quantum interference device (SQUID) loop, which is then threaded by a localized magnetic flux controlled via a dedicated microwave transmission line known as the Z-control line (or flux line).

3.3 Qubit Architecture

Consider a system of two capacitively coupled transmon qubits, one fixed-frequency (QB2) and the other frequency tunable (QB1). The simplified dispersive Jaynes-Cummings Hamiltonian for this system can be modeled as:

$$H = \sum_{i=1,2} \left(\omega_i a_i^\dagger a_i + \frac{\alpha_i}{2} a_i^\dagger a_i^\dagger a_i a_i \right) + g(a_1^\dagger + a_1)(a_2^\dagger + a_2),$$

where g denotes the coupling strength between the two qubits, a_i creation operator for qubit i, and α_i the anharmonicity of qubit i.

3.4 Adiabatic Implementation

To implement the gate, the frequency of Qubit 1 is swept along a prescribed trajectory $\omega_1(t)$ toward the avoided crossing, held or continuously varied for a duration τ_d, and then returned to its idle parking frequency. In this process, $|11\rangle$ obtains/loses more phases than the other computational states, due to its stronger coupling to $|02\rangle$.

By following [2, 12] we review the theory behind the adiabatic implementaion of CZ gate and the derivation of leakage formula Eq. (23).

Theorem (Adiabatic Theorem)**.** *If a quantum system is initially prepared in a non-degenerate energy eigenstate of a time-dependent Hamiltonian, and if that Hamiltonian is varied sufficiently slowly, the system will evolve continuously to remain in the corresponding instantaneous eigenstate of the evolving Hamiltonian up to an accumulated phase factor, given that there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum.*

Consider a two level system with the following Hamiltonian

$$H(t) = \varepsilon(t)\sigma_z + \Delta\sigma_x.$$

According to the adiabatic theorem, if we prepare the state $|0\rangle$ and slowly vary the Hamiltonian and then retrace the change back, the resulting state is $|0\rangle$ with phase accumulation. The instantaneous eigenstates are

$$|\psi_-\rangle = \begin{bmatrix} -\sin(\theta(t)/2) \\ \cos(\theta(t)/2) \end{bmatrix} \quad \text{and} \quad |\psi_+\rangle = \begin{bmatrix} \cos(\theta(t)/2) \\ \sin(\theta(t)/2) \end{bmatrix}, \tag{19}$$

with eigenenergies

$$E_- = -\frac{1}{2}\sqrt{\varepsilon(t)^2 + \Delta^2} \quad \text{and} \quad E_+ = \frac{1}{2}\sqrt{\varepsilon(t)^2 + \Delta^2} \tag{20}$$

where

$$\theta(t) = \arctan\frac{\Delta}{\varepsilon(t)}. \tag{21}$$

We denote the energy difference of these two instantaneous eigenstates as

$$\omega(t) = \sqrt{\varepsilon(t)^2 + \Delta^2} \tag{22}$$

3.4.1 Calculation of leakage error

Consider a state

$$|\psi\rangle = \alpha |\psi_-\rangle + \beta |\psi_+\rangle = \alpha_0 |0\rangle + \beta_0 |1\rangle .,$$

with rotational change of coefficients

$$\alpha = \alpha_0 \cos\frac{\theta}{2} + \beta_0 \sin\frac{\theta}{2}$$
$$\beta = \beta_0 \cos\frac{\theta}{2} - \alpha_0 \sin\frac{\theta}{2}$$

Differentiating, we have

$$\dot{\alpha} = \frac{-i\omega\alpha + \beta\dot{\theta}}{2}$$
$$\dot{\beta} = \frac{i\omega\beta - \alpha\dot{\theta}}{2}$$

Write $\alpha = \cos\Theta/2$ and $\beta = e^{i\phi}\sin\Theta/2$, they form the "angle":

$$\alpha^*\beta = \frac{\sin\Theta e^{i\phi}}{2}$$

Calculating the time derivative of $\alpha^*\beta$ finds us

$$\frac{d}{dt}(\sin\Theta e^{i\phi}) = i\omega\sin\Theta e^{i\phi} \pm \dot{\theta}\cos\Theta$$

Plugging $\phi = \phi_{\text{initial}} + \int_0^t \omega(t')\,\mathrm{d}t'$ and a bit of tinkering yields

$$P_e \approx \frac{1}{4}\left|\int \frac{\mathrm{d}\theta}{\mathrm{d}t} e^{-i\int \omega(t')\,\mathrm{d}t'}\right|^2 \tag{23}$$

3.4.2 Phase accumulation

According to the adiabatic theorem, if the sweep rate is sufficiently slow compared to the energy gap at the avoided crossing, the system will remain in its instantaneous eigenstate. The dynamic frequency excursion imparts a continuous deformation on the $|11\rangle$ state, causing it to hybridize temporarily with $|02\rangle$ and acquire a dynamical phase. The total accumulated phase Φ_{11} on the $|11\rangle$ state is the time integral of its instantaneous eigenenergy $E_{11}(t)$:

$$\Phi_{11} = -\frac{1}{\hbar}\int_0^{\tau} E_{11}(t)dt$$

Simultaneously, single-qubit dynamical phases Φ_{01} and Φ_{10} are acquired. The conditional phase ϕ_{cond} accumulated over the gate duration is calculated by isolating the two-qubit interaction phase from the single-qubit contributions:

$$\phi_{\text{cond}} = \Phi_{11} - \Phi_{01} - \Phi_{10} + \Phi_{00}$$

The gate is successful when the pulse amplitude and duration are calibrated such that $\phi_{\text{cond}} = \pi \pmod{2\pi}$.

3.4.3 Nonlinear time transformation

By applying a time transformation, we can simplify the leakage formula Eq. (23). Let the time variable of this new frame be τ. Set

$$\Delta \mathrm{d}\tau = \omega(t)\mathrm{d}t.$$

then

$$\mathrm{d}t = \frac{\Delta}{\omega(t)}\mathrm{d}\tau = \sin\theta(t)\mathrm{d}\tau. \tag{24}$$

In integral form,

$$t(\tau) = \int_0^\tau \sin\tilde{\theta}(\tau')\,\mathrm{d}\tau' \tag{25}$$

The virtue of this transformation is clear if we plug Eq. (24) into Eq. (23)

$$P_e = \frac{1}{4}\left|\int_0^\tau \frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}\tau}\,e^{-i\Delta\tau}\,\mathrm{d}\tau\right|^2 \tag{26}$$

3.5 Slepian pulses

In the context of pulse waveform optimization, the Slepian pulses, frequently referred to as discrete prolate spheroidal sequences (DPSSs), represent a family of orthogonal pulses specifically formulated to solve the problem of maximal energy concentration in both the time and frequency domains. Because Heisenberg's uncertainty principle dictates that a pulse cannot be strictly bounded in both time and frequency simultaneously, Slepian, Landau, and Pollack developed these sequences to answer a fundamental optimization question: how to optimally concentrate energy in one domain when the pulse is strictly confined in the opposing domain.

To define this optimization rigorously, consider a discrete-time pulse of finite length, denoted as $x[n]$ for $n = 0, 1, \ldots, N-1$. This pulse is constrained to possess finite energy E, defined as:

$$E = \sum_{n=0}^{N-1} |x[n]|^2 < \infty$$

The frequency characteristics of the pulse are analyzed using its discrete-time Fourier transform:

$$X(e^{i\omega}) = \sum_{n=0}^{N-1} x[n]e^{-i\omega n}$$

The objective is to maximize the fraction of the total pulse energy that is contained within a specific frequency band $[-W, W]$, where $0 < W < \pi$. This fractional energy concentration is represented by the ratio $\lambda \in [0, 1]$:

$$\lambda = \frac{\int_{-W}^{W} |X(e^{i\omega})|^2 d\omega}{\int_{-\pi}^{\pi} |X(e^{i\omega})|^2 d\omega}$$

Therefore, the core optimization problem is to identify the specific sequence $x[n]$ of length N that maximizes this eigenvalue λ.3. The Eigenvalue SolutionThe sequences that solve this optimization problem, denoted as $v_n^{(k)}(N, W)$ where $k = 0, 1, \ldots, N-1$ represents the pulse order, are the Slepian pulses. Here, N is the pulse length and W parameterizes the mainlobe width.These pulses are derived as the real solutions to the following system of equations:

$$\sum_{m=0}^{N-1} \frac{\sin 2\pi W(n-m)}{\pi(n-m)} v_m^{(k)}(N, W) = \lambda_k(N, W) v_n^{(k)}(N, W)$$

For the singularity condition where $n = m$, the fraction simplifies to $2W$.This system can be elegantly expressed as the matrix eigenvalue problem $\mathbf{A}\mathbf{v}^{(k)} = \lambda_k \mathbf{v}^{(k)}$.The matrix elements are defined as $A_{n,m} = \frac{\sin 2\pi W(n-m)}{\pi(n-m)}$.The eigenvector is $\mathbf{v}^{(k)} = [v_0^{(k)}(N, W), v_1^{(k)}(N, W), \ldots, v_{N-1}^{(k)}(N, W)]^T$.4. Characteristics and Practical AdvantagesSolving the matrix equation yields N distinct eigenvalues, which are conventionally ranked in descending order: $1 > \lambda_0 > \lambda_1 > \cdots > \lambda_{N-1} > 0$.The sequence $v_n^{(0)}(N, W)$, associated with the strictly largest eigenvalue λ_0, is designated as the first Slepian pulse.Each subsequent higher-order pulse maximizes the energy ratio λ while maintaining strict orthogonality to all preceding Slepian pulses.When evaluating their performance against simpler shapes like rectangular or raised cosine pulses, Slepian pulses are distinguished by possessing both a relatively small mainlobe width and low sidelobe amplitude.This dual efficiency makes them an exceptionally strong candidate for control applications where a precise compromise between mainlobe width and sidelobe suppression is required.

Chapter 4 Proposed Waveform Optimization Techniques

In this section, we investigate methods to mitigate leakage errors in X and CZ gates, beginning with the X gate. Our approach builds on the DRAG-1 scheme which conventionally relies on a seed function to generate a control pulse to minimize leakage from $|0\rangle$ to $|2\rangle$. This conventional method, however, has a disadvantage that it's hard to control the resulting pulse shape. We establish a method to construct DRAG-1 pulse bypassing the seed function method and synthesize a pulse which can achieve a shorter gate duration.

4.1 Note on Y DRAG correction

Aside from the X control pulse, we utilize the DRAG pulses 11 and 12. We demonstrate that the analytical expression for the Y control pulse remains valid even in the non-perturbative regime—a derivation that, to the best of our knowledge, has not been previously reported in the literature. To illustrate this, we recalculate the system Hamiltonian without relying on perturbative assumptions. We introduce a unitary frame transformation V, defined as:

$$V = \exp\left[-i\lambda_2 \frac{\Omega_x}{2\Delta_2}\hat{\sigma}^y_{1,2}\right] = \cos(\phi) - i\hat{\sigma}^y_{1,2}\sin(\phi) \tag{27}$$

where $\phi = \lambda_2 \frac{\Omega_x}{2\Delta_2}$, and apply it to our qudit Hamiltonian in Eq. (7). The $|1\rangle\langle 2|$ component of the resulting Hamiltonian is

$$-\sin(2\phi)\frac{\delta(t)+\Delta}{2} + \cos(2\phi)\frac{\lambda\Omega_x}{2} - i\frac{\lambda}{2}\left(\frac{\Omega'_x(t)}{\Delta} + \Omega_y(t)\right).$$

Substituting Ω_y as defined in Eq. (11) demonstrates that this off-diagonal component is still effectively suppressed. Consequently, we retain the standard DRAG correction for the Y control pulse.

4.2 DRAG-1

Although in 2.6, the DRAG-1 pulse is created from the seed pulse waveform Ω_1 through Eq. (14), now we reverse this process. First we solve Eq. (14) for $\Omega_1(t)$.

We can rewrite Eq. (14) as a second-order linear oridnary differential equation (ODE):

$$\Omega_x^2(t) = \left(1 + \frac{D^2}{\Delta_2^2}\right)\Omega_1^2(t). \tag{28}$$

Recognizing this as a driven harmonic oscillator equation, the general solution is

$$\Omega_1^2(t) = \int_0^t \left[\frac{1}{\Delta_2}\sin(\Delta_2(t-\tau))\right][\Delta_2^2\Omega_x(\tau)^2]\,\mathrm{d}\tau + A\cos(\Delta_2 t + \phi). \tag{29}$$

The first term represents the particular solution to the ODE in Eq. (28). The verification can be done with the Leibniz rule [1], which governs the differentiation under the integral sign:

$$\frac{d}{dt}\int_{a(t)}^{b(t)} F(x,t)dx = F(b(t),t)\cdot b'(t) - F(a(t),t)\cdot a'(t) + \int_{a(t)}^{b(t)} \frac{\partial}{\partial t}F(x,t)dx. \tag{30}$$

Recall we have three boundary conditions on our debt:

1. $\Omega_x = 0$ at both ends ($t = 0,\ T$): This ensures compatibility with the SWT and enforces boundary continuity for the X-pulse, mitigating spectral leakage.
2. $\dot{\Omega}_x = 0$ at both ends ($t = 0,\ T$): This guarantees a smooth envelope derivative, maintaining essential boundary continuity for the corresponding Y-pulse.
3. $\Omega_1 = 0$ at both ends ($t = 0, T$): This further enforces strict SWT compatibility at the start and completion of the gate.

Applying the third boundary condition at the initial time $t = 0$ immediately eliminates the homogeneous component, yielding $A = 0$ in Eq. (29) The constraint at $t = T$ translates to

$$\left(\Omega_1^2(T)\right) = 0 = \int_0^T \left[\frac{1}{\Delta_2}\sin(\Delta_2(T-\tau))\right][\Delta_2^2\Omega_x(\tau)^2]\,\mathrm{d}\tau,$$

differentiating both sides we get

$$\left(\Omega_1(T)\dot{\Omega}_1(T)\right) = 0 = \int_0^T [\cos(\Delta_2(T-\tau))]\,[\Delta_2^2\Omega_x(\tau)^2]\,\mathrm{d}\tau.$$

These two constraints combined implies

$$\int_{-\infty}^{\infty}[\Delta_2^2\Omega_x(\tau)^2]e^{-i\Delta\tau}\,\mathrm{d}\tau.$$

Hence, The boundary condition at $t = T$ for smoothness translates to the null fourier transform condition,

$$\int_0^T \Omega_x(\tau)^2 e^{-i\Delta_2\tau} d\tau = 0. \tag{31}$$

4.2.1 Pulse Design

Although Jesus et al. [8] proposed a pulse design method for Ω_x by way of Ω_1, and investigated an optimal pulse waveform therein, we can directly design a DRAG-1 pulse through Eq. (31). Now we consider pushing the gate time limit for maintaining the perturbative approximation of Schrieffer-Wolff transformation. In it, The problem of early truncation will be somewhat relieved by keeping $\max|\Omega_x|$ small. We combine delayed leakage reduction technique described in [14] and function design techniques to obtain such pulses.

Let $t_d = \pi/\Delta$. Then $f(t) + f(t - t_d)$'s Fourier transform is computed as

$$\mathcal{F}[f(t) + f(t - t_d)](\Delta) = \mathcal{F}[f](\Delta) + e^{i\Delta t_d}\mathcal{F}[f](\Delta) = 0.$$

This suggests that we find a function $g(t)$ on $[-1, 1]$ that satisfies the following property:

1. g is anti-symmetric
2. g is monotonically increasing
3. $g(1) = 1/2$
4. g's first and second derivative vanishes at $g(-1)$.

g represents the function on $[0, t_r]$. On obtaining such a g, we construct Ω_x^2 as

$$\Omega_x^2(t) = \begin{cases} A\left(g(2t/t_r - 1) - g(-1)\right) & (0 < t < t_r) \\ A & (t_r < t < t_d) \\ A\left(g(2(t_r + t_d - t)/t_r - 1) - g(-1)\right) & (t_d < t < t_d + t_r) \end{cases}$$

where A is the amplitude determined by integrating Ω_x^2 according to Eq. (15).

There are mutliple choices for such g.

1. $\frac{1}{2}x + \frac{1}{2\pi}\sin(\pi x)$
2. $3\pi x + 4\sin(\pi x) + \frac{1}{2}\sin(2\pi x)$

4.2.2 g design

Let h be a positive symmetric function on $[-1, 1]$ that vanishes at both ends. Put $g' = h^2$. This ensures that the condition 4 holds as $g'' = 2hh' = 0$ and it is easy to check that the other condtions 1-3 are also satisfied.

4.3 Optimization by Riccati engineering

As we've seen in 2.7, NPAD has an obvious issue that the Hamiltonian is assumed to be time-independent, and in Ref. [11], the author only applies this method to perturbative cases, in which the method reduces to recursive Schrieffer-Wolff transformations.

We outline the newly proposed procedure to handle non-perturbative cases. Applying to the qudit Hamiltonian 6 with $N_{\max} = 2$, the Riccati equation becomes

$$i\dot{x}_1 = (2\delta + \Delta)x_1 - \frac{1}{2}(\Omega_x + i\Omega_y)x_2 - \frac{\lambda}{2}(\Omega_x - i\Omega_y)x_1x_2,$$

$$i\dot{x}_2 = \frac{\lambda}{2}(\Omega_x + i\Omega_y) + (2\delta + \Delta)x_2 - \frac{1}{2}(\Omega_x - i\Omega_y)x_1 - \frac{\lambda}{2}(\Omega_x - i\Omega_y)x_2^2.$$

If we define

$$A = -\frac{1}{2}(x_2^2 + \lambda x_1) \tag{32}$$

$$B = \frac{1}{2}x_1^2 \tag{33}$$

$$C = i(\dot{x_1}x_2 - x_1\dot{x_2}) \tag{34}$$

Then Ω_x, Ω_y, and $\delta(t)$ can be expressed

$$\tilde{\Omega} = \Omega_x + i\Omega_y = \frac{A^\star C - BC^\star}{|A|^2 - |B|^2}, \tag{35}$$

$$\delta(t) = \frac{1}{2}\left(\frac{i\dot{x}_1 + \frac{1}{2}x_2\tilde{\Omega} + \frac{1}{2}\lambda x_1 x_2 \tilde{\Omega}^*}{x_1} - \Delta_2\right). \tag{36}$$

4.3.1 Variational method

One way to utilize Eq. (35) and Eq. (36) is to determine x_1 and x_2 first and solve numerically for $\tilde{\Omega}$ and δ. There's multiple difficulties with this approach. Since the Riccati equation Eq. (**??**) is not linear, adjust the pulse area of Ω_x

is difficult. To overcome this issue, we first derive the and then In this study, Riccati engineering method is utilized as an optimization method, To preserve the X gate condition Eq. (**??**),

$$\delta\Theta = \int_0^T \sum_{k=1}^{2} \left(\frac{\partial \Omega_x}{\partial x_k} \delta x_k + \frac{\partial \Omega_x}{\partial \dot{x}_k} \delta \dot{x}_k + \text{c.c.} \right) dt \tag{37}$$

must vanish. ($\frac{\partial}{\partial x_i}$ denotes Wirtinger derivatives.) Integral by parts transforms the above equation into

$$\delta\Theta = \left[\sum_{k=1}^{2} \left(P_k(T) \delta x_k(T) + P_k^*(T) \delta x_k^*(T) \right) \right] + \int_0^T \sum_{k=1}^{2} \left(G_k(t) \delta x_k(t) + G_k^*(t) \delta x_k^*(t) \right) dt. \tag{38}$$

Note that $\delta x_k(0) = 0$.

$$G_k(t) = \frac{\partial \Omega_x}{\partial x_k} - \frac{d}{dt} \left(\frac{\partial \Omega_x}{\partial \dot{x}_k} \right)$$

$$P_1(t) = \frac{\partial \Omega_x}{\partial \dot{x}_1} = i \frac{A^* - B^*}{2(|A|^2 - |B|^2)} x_2$$

$$P_2(t) = \frac{\partial \Omega_x}{\partial \dot{x}_2} = -i \frac{A^* - B^*}{2(|A|^2 - |B|^2)} x_1$$

4.4 CZ gate

In this section, for brevity, the length of pulses are assumed to be N, an odd number. As seen in 3.4.1, the leakage error is approximately expressed as $\frac{1}{4}|G(\Delta)|^2$. In the following we propose several methods to mitigate this error.

4.4.1 anti-symmetric Chebyshev pulse

Maintaining a specified sidelobe level, Chebyshev pulse is the signal that minimizes the mainlobe width

Mathematically, a chebyshev pulse g_{ch} is a discrete time pulse defined as

$$g_{\text{ch}}[n] = \frac{1}{N} \left[1 + 2r \sum_{k=0}^{(N-1)/2} (-1)^k T_{N-1} \left(x_0 \cos \frac{\pi k}{N} \right) \cos \left(\frac{2\pi k}{L} \left(n + \frac{1}{2} \right) \right) \right],$$

where $r = \frac{1}{T_{N-1}(x_0)}$ corresponds to the suppressed sidelobe level. We call its anti-symmetrization

$$g_{ach}[n] = \begin{cases} g_{ch}[n] & n < (N+1)/2 \\ 0 & n = (N+1)/2 \\ -g_{ch}[N-n] & n > (N+1)/2 \end{cases}$$

anti-symmetric Chebyshev pulse g_{ach}.

4.4.2 Correctional Derivative pulse

To reduce the fourier transform at Δ, recall that

$$\mathcal{F}[f'](\Delta) = i\omega \mathcal{F}[f](\Delta).$$

Hence, by adding the second derivative to f, we can reduce the fourier transform to 0,

$$\mathcal{F}[(1 + D^2/\Delta^2)f](\Delta) = 0.$$

This method is only applicable to pulses whose second derivative is anyti-symmetric, such as anti-symmetric Chebyshev, to maintain the anti-symmetric property after adding the second derivative.

4.4.3 Combining two pulses

Combining two pulses of different origins

4.4.4 Redefining the Slepian eigenvalue problem

The (second) Slepian pulse is the solution of the second largest eigenvalue to the following problem

$$\frac{\int_{-W}^{W} |G_{\mathrm{DTFT}}(\omega)|^2 d\omega}{\int_{-\pi}^{\pi} |G_{\mathrm{DTFT}}(\omega)|^2 d\omega} = \frac{g^T A_{m,n} g}{g^T g} \tag{39}$$

where $A_{m,n} = \mathrm{sinc}(\pi(m-n)W) = \dfrac{\sin(\pi(m-n)W)}{\pi(m-n)W}$. (39) is the eigenvalue problem of the matrix $A_{m,n}$. It was Slepian who found a commuting matrix for $A_{m,n}$ and solev teh problem analyitcally.

Close investigation of the Eq. (**??**)

For the targetted range of $W_1 \leq \dfrac{T}{N}\Delta \leq W_2$

$$\text{minimize } \frac{\int_{W_1}^{W_2} |G_{\mathrm{DTFT}}(\omega)|^2 d\omega}{\int_{W_1}^{W_2} |G_{\mathrm{DTFT}}(\omega)|^2 d\omega} = \frac{\tilde{g}^T (A_{\mathrm{rect}}(W_2) - A_{\mathrm{rect}}(W_1))\tilde{g}}{\tilde{g}[0]^2 + \cdots + \tilde{g}[N]^2}$$

To not elongate gate time we impose the following condition, $\tilde{g}$ should be anti-symmetric and non-negative on the first half.

- $\tilde{g}[s] = \tilde{g}[N-1-s]$
- $\tilde{g}[s] \geq 0 \quad \text{for } 0 \leq s \leq L-1$

Put

$$B = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ 0 & \cdots & 0 \\ & & -1 \\ & \iddots & \\ -1 & & \end{pmatrix}, \quad C = B^T A B$$

$$\begin{pmatrix} \tilde{g}[0] \\ \vdots \\ \tilde{g}[N] \end{pmatrix} = B \begin{pmatrix} h[0] \\ \vdots \\ h[L-1] \end{pmatrix}$$

Then

$$\text{maximize} \quad -\frac{h^T C h}{h^T h}$$

is the problem to solve.

4.4.5 Sequential Quadratic Programming

FAST [7] Given an ansatz and an evaluation point Δ, the algorithm goes as follows:

Algorithm 1 Variational Quantum Circuit Optimization

Require: Target unitary U, Error tolerance ϵ

Ensure: Optimized parameters $\boldsymbol{\theta}$

Initialize $\boldsymbol{\theta} \leftarrow$ random values in $[0, 2\pi]$

Set learning rate $\alpha \leftarrow 0.01$

while cost function $\mathcal{L}(\boldsymbol{\theta}) > \epsilon$ **do**

  Compute gradient $\nabla\mathcal{L}(\boldsymbol{\theta})$ using parameter-shift rule

  if $\|\nabla\mathcal{L}(\boldsymbol{\theta})\| <$ threshold **then**

    break ▷ Early stopping criteria

  end if

  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha\nabla\mathcal{L}(\boldsymbol{\theta})$

end while

return $\boldsymbol{\theta}$

Chapter 5 Numerical Results and Discussions

For quantum simulation, we used QuTiP5 [9] software.

5.1 X gate

5.1.1 Simulation settings

Simulation has been conducted with the anharmonicity $\Delta/2\pi = -0.225$ GHz. The Hamiltonian used is Eq. (7) with $N_{max} = 3$. To see the limitation of perturbative method, the performance of the DRAG-1 pulse implementing the X gate is simulated. The is created with $\Omega_1(t) = \Omega_0 \sin^3(\pi t/T)$, which gives

$$\Omega(x) =$$

The maximum value of which is

5.1.2 fidelity

In the case of dynamics without dissipative effects, the gate infidelity is defined as $\mathcal{E}(\hat{U}_Q) = 1 - \mathcal{F}[\hat{U}_Q]$ where $\mathcal{F}[\hat{U}_Q]$ is the gate fidelity defined as [13]

$$\mathcal{F}[\hat{U}_Q] = \frac{\mathrm{Tr}[\hat{U}_Q \hat{U}_Q^\dagger]}{d(d+1)} + \frac{\left|\mathrm{Tr}[\hat{U}_Q \hat{U}_I^\dagger]\right|^2}{d(d+1)}, \tag{40}$$

5.2 CZ gate

5.2.1 Simulation

As in the previous research [2], we converted the pulse in the following manner:

$$\tilde{g} \to \tilde{\theta} \to x \to y$$

Simulation conditions are

5.2.2 Anti-symmetric Chebyshev pulse

Created with with the parameter $r = 0.000238$.

5.2.3 Leakage

ε

5.2.4 Minimal Gate time

Once the pulse shape $\tilde{g}$ determined, its integration $\tilde{\theta}$ is decided up to the amplitude factor $A(\tau_d)$, $(0 \le A(\tau_d) \le 1)$

$$\tilde{\theta} = \tilde{\theta}^n(\tau/\tau_d)A(\tau_d),$$

where $\tilde{\theta}^n$ is the normalized $\tilde{\theta}$ that reaches $\epsilon(t_{mid}) = 0$ during avoided crossing.

$$\begin{aligned} t(\tau) &= \int_0^\tau \sin \tilde{\theta}(\tau') \, \mathrm{d}\tau' \\ &= \tau_d \int_0^{\tau/\tau_d} \sin \left[A(\tau_d) \tilde{\theta}^n(\tau') \right] \, \mathrm{d}\tau'. \end{aligned}$$

During this flux trajectory, the states $|00\rangle$, $|01\rangle$, and $|10\rangle$ remain largely isolated from any crossings. Therefore, the phases they accumulate are purely their standard, uncoupled dynamical phases:$\phi_{00} = \phi_{00}^{\text{uncoupled}} \phi_{01} = \phi_{01}^{\text{uncoupled}} \phi_{10} = \phi_{10}^{\text{uncoupled}}$ Therefore, gate duration in the lab frame is $t_d = \tau_d \int_0^1 \sin \left[A(\tau_d) \tilde{\theta}^n(\tau') \right] \, \mathrm{d}\tau'$. t_d is decided upon the amplitude $A(\tau_d)$ and the initial pulse shape. There are two types of phases: Berry phases and dynamical phases. Since the trajectory that the instantaneous $|11\rangle$ traces through the operation is a 1D path, thus enclosing zero area, it produces zero Berry phase. In formula,

$$\phi = \int \Delta E \, \mathrm{d}t = -\int \frac{\Delta}{2} \tan \frac{\theta(t)}{2} \, \mathrm{d}t \tag{41}$$

Since $\tan(\theta/2) = \dfrac{\tan(\theta)}{1 + \sqrt{1 + \tan(\theta)^2}}$

As you see in the graph of $\epsilon(t)$ Since $0 \leq \tan \dfrac{\theta(t)}{2} \leq 1$, the gate duration T satisfies $T \geq \dfrac{2\pi}{\Delta}$.

is connected through

$$\omega_1(\Phi_{\text{ext}}) = \frac{1}{\hbar} \left(\sqrt{8E_J E_C} \sqrt[4]{d^2 + (1 - d^2) \cos^2(\pi \frac{\Phi_{\text{ext}}}{\Phi_0})} - E_C \right), \tag{42}$$

Required slew rate of DAC is

5.2.5 Combination of two pulses

Interestingly enough, their arguments are almost equal defying the possibility of mixing them together to gain a pulse with lower leakage.

5.2.6 Robustness against gate time

The reason behind the fact that the Slepian pulse has been used for so long days is its robustness against gate duration.

Chapter 6 Conclusion

Riccati engineering is applied to optimize an existing pulse, if we could simplify the Hamiltonian somehow, could be available.

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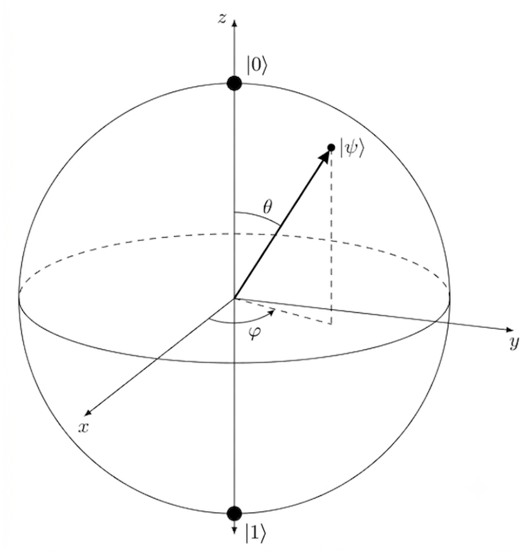

Figure 1: $\alpha\beta$

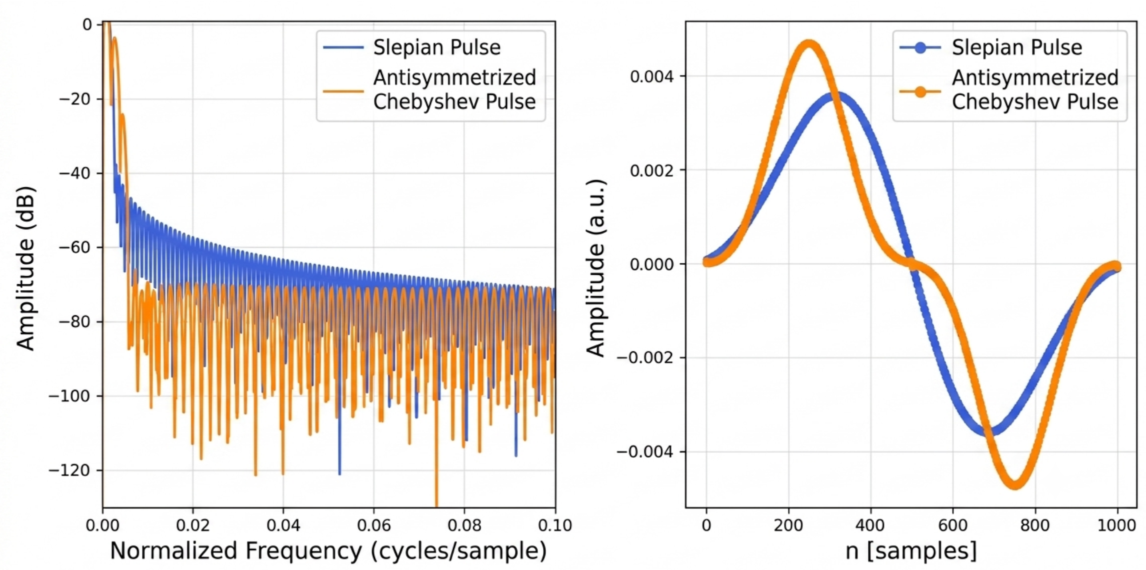

Figure 2: The parameter r is chosen so that sidelobe height is 40 dB lower than the first slepian sidelobe height.

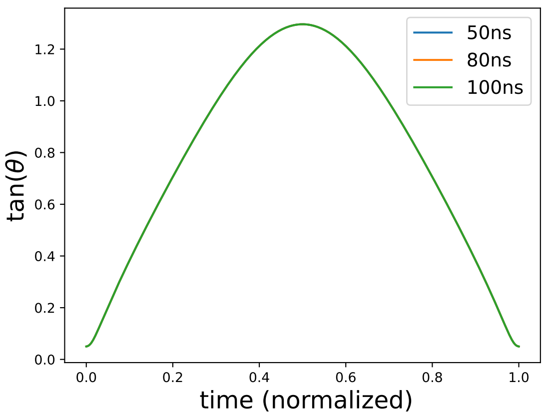

Figure 3: The relation between τ and t for the Slepian pulse with $N = 50$, $\Delta = 0.8$, and $T = 100$. The three graphs are overlapping each other due to almost the same $A(\tau_d)$.